

# Sina Mousavi Rad, Ph.D.

in LinkedIn

With Work Authorization

📍 Willing to Relocate

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☎ (572) 220-1060

## SUMMARY

Civil Engineer with a Ph.D. and expertise in pavement engineering, seeking a full-time position in Pavement work, with experience in asphalt pavement materials, pavement design, performance-based testing & mix design, pavement condition evaluation, pavement management, and data-driven analysis.

## EMPLOYMENT

- **Research Assistant | Oklahoma State University | USA** 2023 – 2026
  - ✓ Conduct Performance Testing of Asphalt Mixtures and Binders & Data Analysis.
  - ✓ Support Research on Asphalt Pavements and Balanced Mix Design (BMD).
- **Research Assistant | Iran University Science and Technology | Iran** 2019 – 2021
  - ✓ Performed Laboratory Testing of Asphalt Mixtures.
  - ✓ Processed and Analyzed Experimental Data for Pavement Materials Research.
- **Road & Pavement Engineer | PARDARAZ Consulting Company | Iran** 2018 – 2019
  - ✓ Conducted Pavement Condition Surveys & Pavement Design.
  - ✓ Performed Field Assessments of Roadway Networks to Support Maintenance Decisions.

## SKILLS & EXPERTISE

- **Data Modeling & Programming:** Python, Machine Learning.
- **Statistical & Data Analysis:** JMP, Excel, R.
- **GIS & Design Software:** ArcGIS, AutoCAD
- **Field Data Collection & Writing Technical Reports**
- **Pavement Design & Engineering:** FAARFIELD, Mn-PAVE, PCI, PAVER, Pavement Condition Evaluation.
- **Asphalt Materials Testing:** Superpave, Asphalt Mixture Performance Testing, Binder Rheological Testing.
- **BMD & Recycled Asphalt Materials**

## EDUCATION

|                            |                                                 |          |             |
|----------------------------|-------------------------------------------------|----------|-------------|
| Ph.D. in Civil Engineering | Oklahoma State University, USA                  | GPA: 4.0 | 2023 – 2026 |
| M.Sc. in Civil Engineering | Iran University of Science and Technology, Iran | GPA: 3.7 | 2018 – 2021 |
| B.Sc. in Civil Engineering | Tehran Azad University, Iran                    | GPA: 3.6 | 2014 – 2018 |

## PROFESSIONAL SERVICE

- **Research Team Coordinator | OSU Asphalt Research Group | USA.** 2024 – 2026
  - ✓ Led Research Team Activities & Managed Laboratory Operations.
- **Research Engineer | USA.** 2023 – 2026
  - ✓ Prepared Technical Reports & Managed DOT-Funded Multi-Year Research Projects.
- **Peer Reviewer | Journal of CBM (Elsevier).** 2022 – 2026
  - ✓ Reviewed Technical Manuscripts Related to pavement engineering.
- **Online Instructor | FaraDars Institute | Iran.** 2021 – 2023
  - ✓ Developed Online Course Content for Pavement Management: PCI and PAVER.
- **Author & Translator | Iran.** 2019 – 2021
  - ✓ Translated the Persian Edition of Route Location and Design by Thomas F. Hickerson.

## CERTIFICATE

- **Fundamentals of Engineering (FE) | Civil** 2026
- **Python for Data Science and Machine Learning Bootcamp** 2024
- **Practical Data Science with MATLAB Specialization** 2022
- **Mastering Bitumen for Better Roads and Innovative Applications** 2022

## ACADEMIC HONORS & AWARDS

- **Winning 1<sup>st</sup> Place for Best Presentation** | Oklahoma Transportation Research Symposium **2026**  
✓ **Project:** Asphalt Mixture Performance & Rheological Evaluation.
- **Winning 2<sup>st</sup> Place for Best Presentation** | Oklahoma Transportation Research Day (OTRD) **2025**  
✓ **Project:** Machine Learning-Based Software for Pavement Cracking Prediction.
- **Winning 1<sup>st</sup> Place for Best Presentation** | Oklahoma Transportation Research Day (OTRD) **2023**  
✓ **Project:** Oklahoma BMD Investigation.
- **Awarded Silver TRL 22<sup>nd</sup>** | National Exhibition of Research and Technology Achievement **2022**  
✓ **Project:** Image Processing to Detect Pavement Cracks & Damages.
- **Awarded Silver TRL 21<sup>nd</sup>** | National Exhibition of Research and Technology Achievement **2021**  
✓ **Project:** A Novel Eco-Friendly Permeable Pavement.

## SELECTED PUBLICATIONS

- **Sina Mousavi Rad** and et al. *"Evaluation of Asphalt Mixture Cracking Resistance and Development of a Machine Learning-Based Application for Cracking Prediction"*. Journal of Construction and Building Materials. **(2025)**.
- M. Olagunju, **Sina Mousavi Rad**, and et al. *"Evaluating the Rutting and Moisture Resistance of Asphalt Mixes in Oklahoma"*. Journal of Transportation Research Record (TRR). **(2025)**.
- **Sina Mousavi Rad** and et al. *"Impact of Binder and Aggregate Type on the Cracking, Rutting, and Moisture Susceptibility of Asphalt Mixes in Oklahoma: A BMD Approach"*. Journal of Materials in Civil Engineering. **(2024)**.
- **Sina Mousavi Rad** and et al. *"The Feasibility of Concurrent Improvement of Pollutants Absorption Ability from Surface Runoff and Mechanical Performance of Porous Asphalt"*. Journal of Materials in Civil Engineering. **(2023)**.
- **Sina Mousavi Rad** and et al. *"Evaluation of Fatigue Performance and Self-Healing Behavior of Nano-ZnO Modified Porous Asphalt Containing RAP"*. Journal of Materials in Civil Engineering. **(2022)**.

## REFERENCES

- **Dr. David Vivanco** | ODOT | Asphalt Branch Manager **DVivanco@odot.org | (405)5224986**
- **Dr. Deb Mishra** | Virginia Tech | Associate Professor **debmishra@vt.edu | (806)2528558**
- **Dr. Joshua Li** | Oklahoma State University | Professor **qiang.li@okstate.edu | (405)7446697**